

# Balireddy Mahalakshmi

Badvel, Andhra Pradesh - 516227 | 7200481646 | balireddymahalakshmi04@gmail.com  
linkedin.com/in/balireddy-mahalakshmi | github.com/Mahalakshmibalireddy21003680

## Career Objective

AI/ML Graduate with a strong foundation in building intelligent systems that solve real-world problems. I specialize in developing deep learning models—like YOLO and LSTMs—and connecting them to live data sources. I have hands-on experience in Python automation, predictive analytics, and deploying solutions using AWS cloud. I'm passionate about making AI practical, scalable, and easy to use in everyday life.

## Technical Skills

- **Programming Languages:** Python, SQL (MySQL), C, Data Structures & Algorithms (DSA).
- **Machine Learning:** Model Development, Deep Learning Architectures, NLP, Data cleaning & preprocessing.
- **Libraries & Frameworks:** TensorFlow, Keras, OpenCV, Pandas, NumPy, Matplotlib.
- **Tools & Cloud:** AWS (Certified), FastAPI, Postman, Git, GitHub, Power BI, VS Code, MS Excel.
- **Soft Skills:** Collaborated with cross functional teams, Documented ML Workflows, communicated findings using data visualization tools

## Professional Experience

**Machine Learning Intern** | FabHost Tech Solutions | Jan 2024 – Mar 2024

- Developed and deployed supervised ML models (Random Forest, XGBoost) using Scikit-learn, performing hyperparameter tuning with GridSearchCV and evaluating performance using Precision, Recall, F1-score, and ROC-AUC metrics.
- Automated **data-cleaning pipelines** by developing custom Python scripts, reducing manual data preparation time by 20% and ensuring high data integrity.
- Streamlined ML workflows by implementing modular code and version control, improving the scalability and reproducibility of team-wide experiments.

## Projects

### Netra: AI-Based Real-Time Assistive Vision System

- Developed a **multi-modal AI pipeline** integrating YOLOv8 for object detection, **EasyOCR** for text recognition, and **MobileNetV2** for high-accuracy currency classification.
- Architected a **Client-Server infrastructure** using **Android (CameraX)** and **FastAPI**, enabling real-time frame capture and high-speed model inference via a modular backend.
- Containerized the **FastAPI backend** using **Docker** and deployed on **AWS EC2** for scalable real-time inference, handling concurrent API requests with **sub-200ms latency**.
- Reduced inference latency by 35% through OpenCV preprocessing and **asynchronous API (Retrofit)** handling to deliver seamless, low-latency audio feedback via Text-to-Speech (TTS).

### IoT-Based Weather Forecasting System

- Architected a **real-time data pipeline** using **ESP32** sensors and **TensorFlow**, achieved 91% forecasting accuracy on 12-month time-series data using LSTM.
- Engineered an **interactive dashboard** by integrating **Google Maps API**, enabling real-time location-based data visualization and sensor mapping.
- Developed **automated alerting workflows** using **IFTTT protocols** and asynchronous data streams to provide instant mobile notifications based on environmental thresholds.

## Education

**Bachelor of Technology (B.Tech) in AI & Machine Learning**

Saveetha Engineering College | Chennai, TN | 2021 – 2025 | **CGPA: 8.0/10**

## Certifications

- AWS Certified Cloud Practitioner | Amazon Web Services (EC2, S3, IAM, Cloudwatch)
- Machine Learning with Python | IBM / Udemy
- Python Programming Certification | FabHost Tech Solutions